One-Time Pad Version of the Vigenère Cipher

35. Write a C program for one-time pad version of the Vigenère cipher. In this scheme, the key is a stream of random numbers between 0 and 26. For example, if the key is 3 19 5 . . . , then the first letter of plaintext is encrypted with a shift of 3 letters, the second with a shift of 19 letters, the third with a shift of 5 letters, and so on.

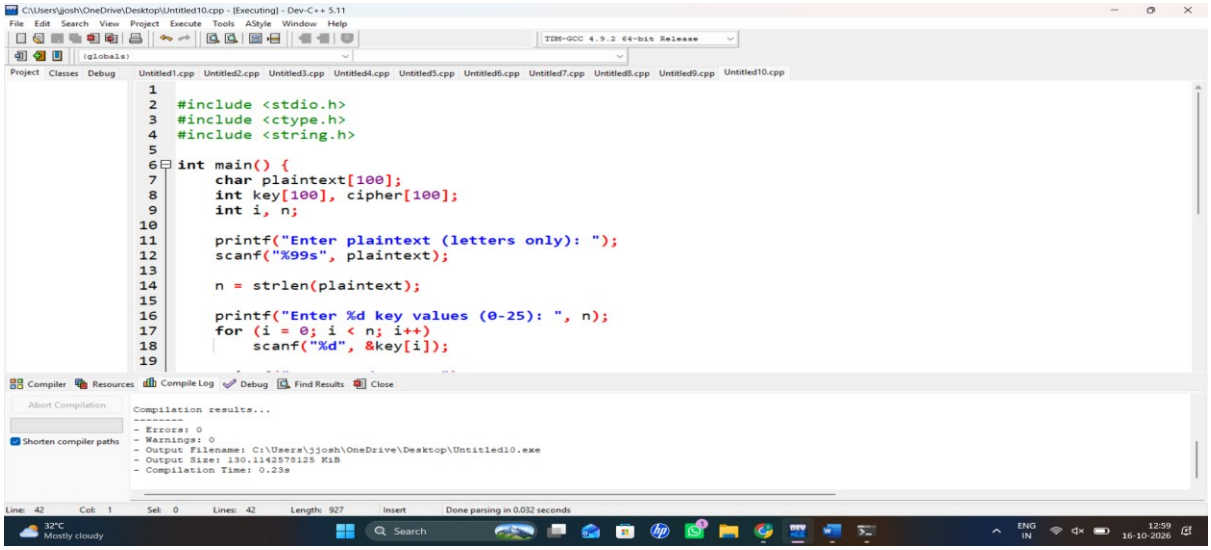
Aim:

To implement the one-time pad version of the Vigenère cipher in C using a sequence of key values between 0 and 25.

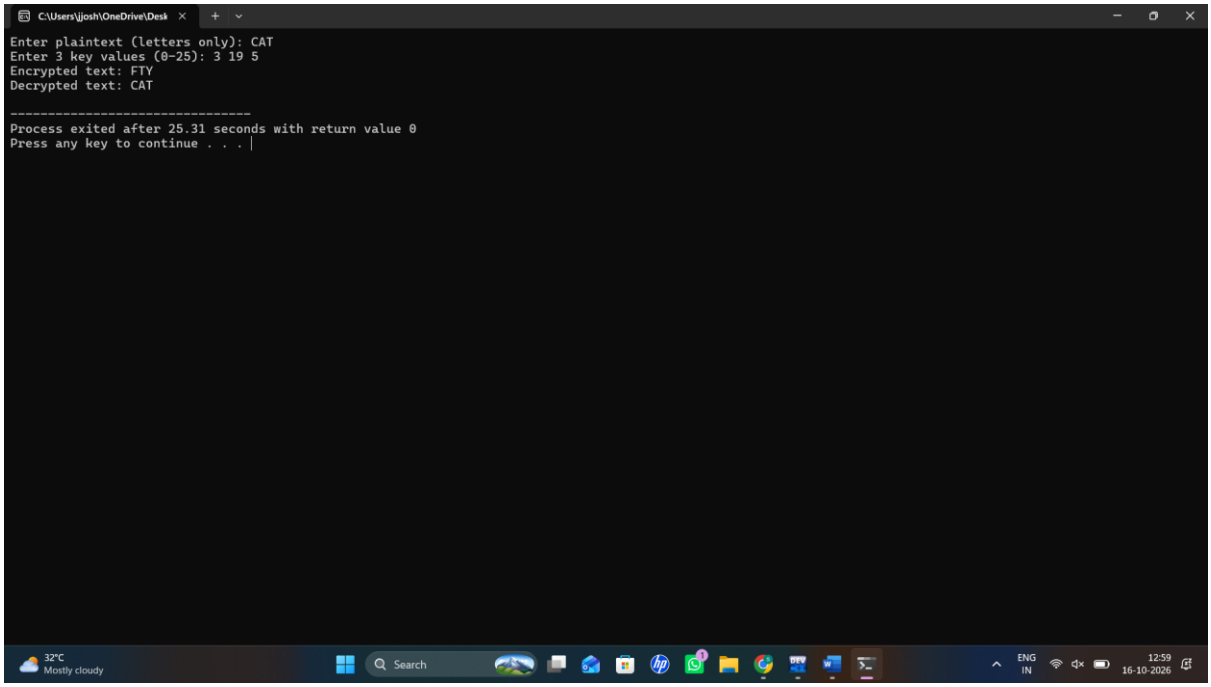
Algorithm:

1. Read the plaintext containing alphabetic characters.
2. Convert each letter into a number from 0 to 25, where A = 0 and Z = 25.
3. Generate or provide a random key value between 0 and 25 for every plaintext character.
4. Encrypt each character using $C_i=(P_i+K_i) \bmod 26$.
5. Decrypt using $P_i=(C_i-K_i+26) \bmod 26$.
6. Display the ciphertext and recovered plaintext.

Program:



Input/Output:



Result:

The one-time pad version of the Vigenère cipher was implemented successfully. The plaintext was encrypted using different key values and recovered through decryption.